Welp, It’s Time to Fess Up: We Got Sunlight All Wrong

Published by
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Estimated reading time
: 14 min
Added on
: 2026-06-18
Address
: https://slate.com/life/2026/06/book-sun-science-defense-sunscreen-uv-light-spf-sunlight.html

Welp, It’s Time to Fess Up: We Got Sunlight All Wrong

The following is an excerpt from Rowan Jacobsen’s forthcoming book, In Defense of Sunlight: The Surprising Science of Sun Exposure*, out from Simon & Schuster June 16. It has been edited and condensed for Slate.*

In the spring of 2025, I became intensely fascinated by a topic that, for the last seven years, had become increasingly present in my mind. That topic, which would soon consume my every waking thought and which I would eventually turn into a book, was sunlight.

The curiosity stemmed from an ongoing observational study of a single subject: me. I live in Vermont. Gorgeous summers, brutal winters. Over the years, I started to notice what a profound effect the different seasons had on me. Basically, winter sucked. This was not simple Seasonal Affective Disorder, a common condition in which people feel low during the winter months, and which is believed to be caused by a dearth of light entering the eyes. I didn’t feel depressed in winter. I felt as if my cells didn’t work.

In summer, my senses were sharper. My thinking was sharper. I ate less but did more, as if nutrients and oxygen were flowing more freely. My skin felt snappier, especially once it got a little color. And I was way happier. Small things seemed deeply pleasing.

And then, when the days grew shorter and the temperature dropped, it all went away. This was not due to lack of exercise. I go for a cross-country ski every decent day of winter. I probably get *more* focused cardio workouts in the snowy months. But it didn’t matter. In winter, my brain and body never felt like they were firing on all cylinders.

It also wasn’t just some inchoate malaise. I could see the difference. In the mirror, I looked a little bit puffy. Despite all the skiing, I’d have a hint of a pouch by late winter. My blood pressure would creep up a few points.

And it wasn’t just me. Most people I knew lost their sharpness in winter. Many got depressed. We all commiserated about the short days. Human beings don’t hibernate like bears or groundhogs, but our bodies were clearly going into some sort of power-saver mode when the sun withdrew.

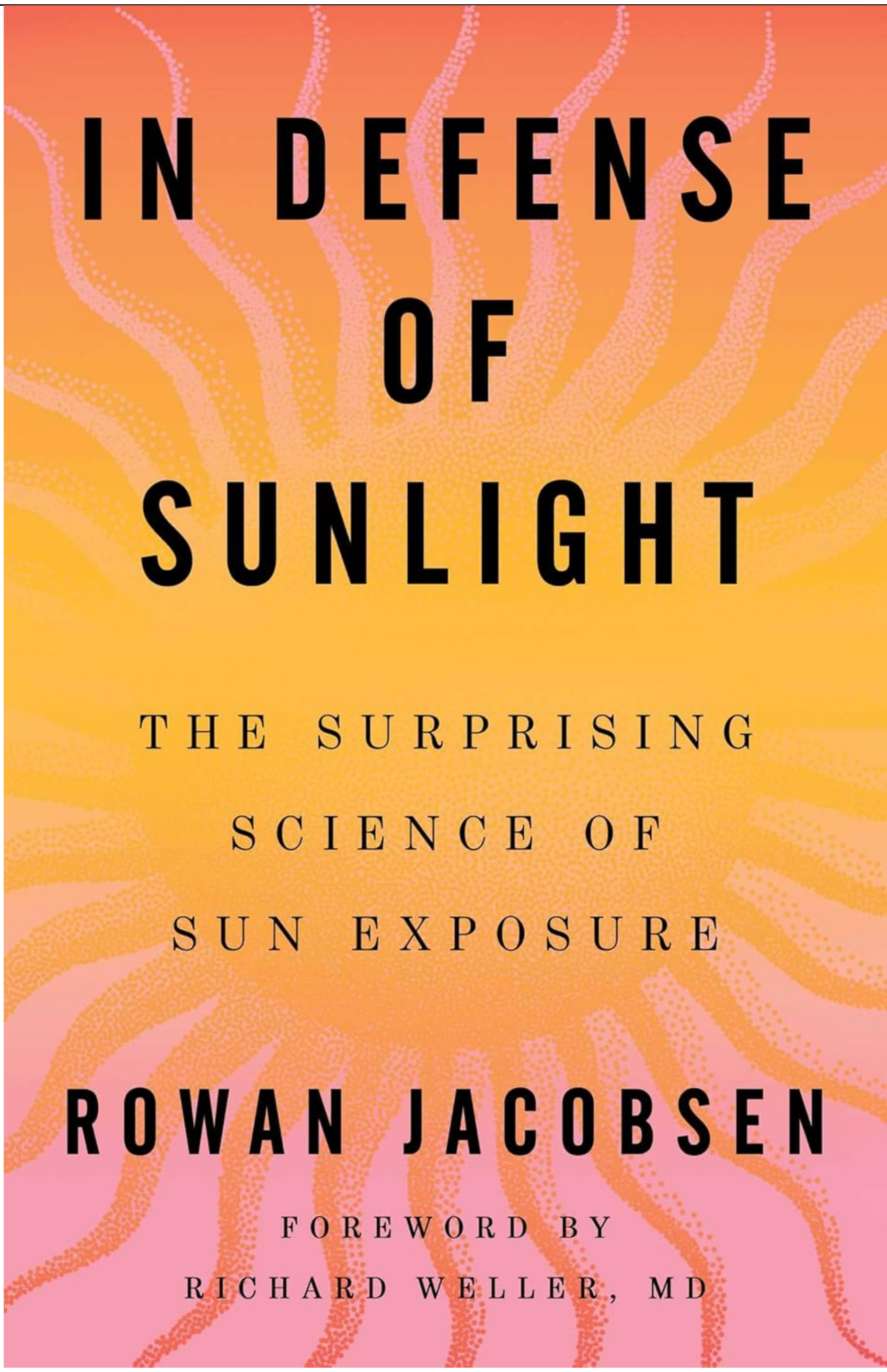
It seemed like a no-brainer that it had something to do with light. A bright spring day after a long cloudy spell was like drinking from a well after crawling through a desert. But I never thought much about the mechanism until 2014, when a team of dermatologists at Harvard Medical School confirmed the sun’s mood-boosting powers. It turns out that when our skin cells are exposed to sunlight, they produce a range of hormones that have profound effects on the body and brain. One triggers the production of melanin, the dark pigment that makes us tan and protects us from burning, and which is also a powerful antioxidant that can mop up the damage caused by the sun’s rays.

Sunlight also initiates the production of cortisol, the hormone of activity. It makes your cells fire faster. You get more alert, you burn more energy, and you feel more focused. People think of cortisol as the “stress hormone” and think it’s a sign of trouble, but cortisol is actually the dealing-with-stress hormone—and stress can, in this case, be anything the world throws at you that requires a response, good or bad. Cortisol is only a problem when it’s chronically overproduced in response to too much stress. I actually think of cortisol as the “in the zone” hormone. When you are in a tennis match, ready to pounce on a weak serve and fire a cross-court winner, that’s cortisol talking.

When you are crushing charades at game night, that's also cortisol. It's the hormone of high performance, of rising to the occasion, and it naturally rises in the morning, but happens faster when sun kisses skin.

A third sun-responsive compound is an endorphin that suffuses the brain with natural opioids, triggering a blissful state. That one was the focus of the Harvard researchers, who found that blocking the opioids produced withdrawal symptoms. In their paper and accompanying press release, they announced that sunlight is literally addictive.

To me, the discovery had profound evolutionary implications. Biological reward systems evolve to encourage beneficial behavior. So the discovery was somewhat mystifying. Like most people, I'd had it drilled into me since childhood that the sun was bad news, the primary cause of skin cancer, and that the only acceptable dosage that should be allowed to touch my unprotected skin was zero. But if we're wired to seek sunlight, there must be a reason.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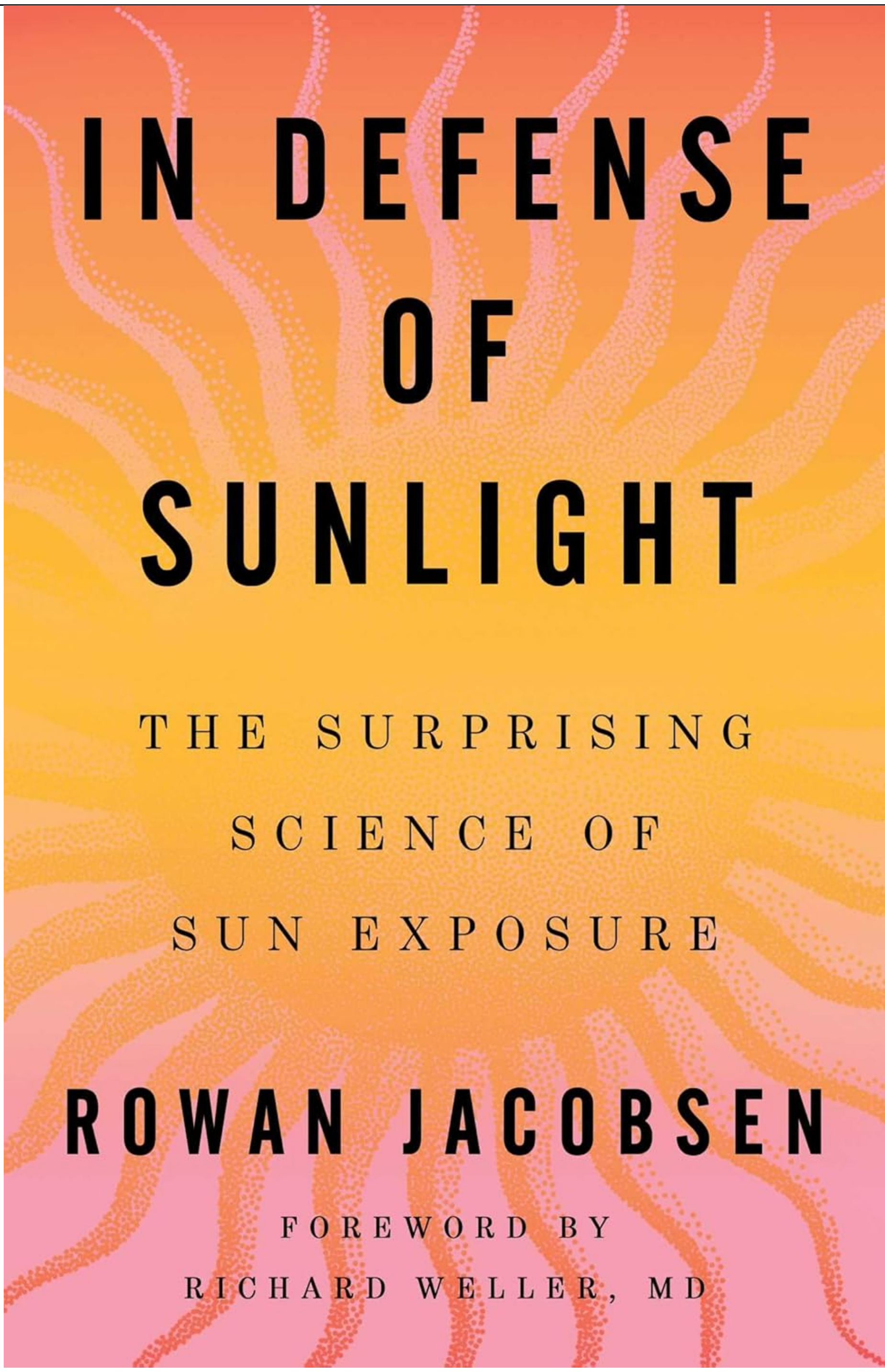

In Defense of Sunlight: The Surprising Science of Sun Exposure

By Rowan Jacobsen. Scribner.

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But that wasn't how the Harvard doctors saw it. They acknowledged that the discovery implied some "potential evolutionary benefit … that reinforces UV-seeking behavior," but they didn't want to touch that one with a 10-foot parasol. Instead, they focused on how this helped explain why it was so hard to break people of their sun habits. There was even talk of treating recreational tanning as a form of drug abuse.

To me, it seemed obvious that we needed to flip the question around. If sun hitting skin in the morning helped energize us for the day ahead, and made us feel great to ensure we'd be right back out there at dawn the next day, might there be other biological benefits we hadn't yet discovered?

More pressingly, what happens to us if sunlight *never* hits our skin? By 2014 we were well on our way to becoming an indoor species, a trend that has only accelerated since then. When we do venture outside, we make sure to block the light. Are we depriving ourselves of something essential? Something our basic biology had been built for? And could part of the modern world's malaise be chalked up to sun deficiency?

I began experimenting a bit. Sure enough, a shot of morning light led to productive, high-energy days. I also began to notice how much better I felt the instant I stepped outside. Winter, summer, clouds, eyes closed, eyes open, it didn't matter, there was always an instantaneous clearing of the cobwebs, a cellular snapping-to. It was as if some nutrient was pouring from the sky.

The story got even more interesting in 2018, when a paradigm-shifting study appeared in Cell, one of the world's most impactful science journals. The researchers, a neurology team from China, had developed a new technology that allowed them to analyze the makeup of individual neurons. A lot of mind-blowing work came out of the technique, but the paper in Cell showed that sunlight hitting skin made neurons work better and improved learning and memory in mice by triggering a release of glutamate, a vital neurotransmitter. The extra glutamate reliably increased brain activity and improved mood and cognition. (This is similar to exercise, which also raises glutamate activity in the brain and relieves depression.) Since then, other studies have confirmed the sun's ability to boost cognition and mood in human beings through multiple pathways.

Here were good reasons why the body might be rewarding sun-seeking behavior with an extra dose of endorphins. Not only did sunlight make you happier, it literally fed your brain. This was a revelation, and it was treated as such in the scientific press. "We are all aware that the knee bone is connected to the thigh bone, but did you know that the brain was connected to the skin?" Science magazine opined in a commentary on the paper. "You'd have to think that beta-endorphin and glutamate are not going to be the end of it (although those two are powerful enough for plenty of effects). One immediately wonders about the well-known effects of sunlight on seasonal depression, among other possibilities."

Two neurologists from Harvard Medical School agreed, penning a commentary in Cell titled "Sunlight Brightens Learning and Memory." "Sunlight alters physiology in complex ways," they wrote, suggesting that the newly discovered mechanism "is likely not the sole mediator of its effects on mood and behavior. … This dataset provides a great starting point to delve into other mechanisms responsible for sunlight-induced changes in behavior." And they didn't disguise their position: "Whereas exposure to excessive levels of sunlight is detrimental to our health, moderate exposure can boost our physical and mental state."

In the world of journalism, I seemed to be the only one who was captivated by the news. And the more I dug around, the more astonished I got. The science consistently showed that people who received ample amounts of sun exposure lived longer, healthier lives than people who didn't. Immunologists were finding that sun deprivation during childhood led to higher rates of autoimmune diseases. Cardiologists were noting lower blood pressure and less diabetes and heart disease in those who got a daily dose of sunshine. Endocrinologists were realizing how many hormones and neurotransmitters were produced when sunlight hit skin. Chronobiologists were seeing accelerated aging in night-shift workers and astronauts. Neurologists were discovering the surprising ability of certain wavelengths to reduce pain and enhance wound healing. And psychologists were learning that sunlight's ability to keep the mind on an even keel was as effective as antidepressants. The skin was turning out to be the organ that set the tone for the functioning of the entire body, and we'd unwittingly been confusing it for decades.

The evidence had been growing stronger as the tools got better. For instance, one massive U.K. study attached light-sensing watches to the wrists of 89,000 volunteers to document their weekly light exposure, then followed the group for eight years to see who died and who didn't.

Those who didn't get enough daylight died far more. People exposed to the most light were 34 percent less likely to expire than people who got less daylight than average. Even after adjusting for everything from exercise and diet to smoking and sleep duration, the sun-getters were still 17 percent less likely to perish.

Why? The short answer is that the human body evolved in an environment of abundant sunlight, and we are designed to make use of that solar energy in ingenious ways that we are finally starting to understand. It's an elegant system that's been fine-tuned over billions of years of existence on a sun-drenched planet.

But it's also one that's been thrown into disarray in recent times with our retreat from the outdoor world. Most Americans spend less than an hour per day outside. One in five virtually never sets foot beyond the built environment. That leaves us trapped in what chronobiologists call biological darkness—weak and weird artificial lighting that is so different from full-spectrum sunlight that our cells can't make use of it.

Cut off from its ancient lodestar, the body loses its way. The brain gets trapped in a never-ending pit of seasonal affective disorder. The ramifications are so alarming that 15 scientists with expertise in these fields recently published a paper titled "Insufficient Sun Exposure Has Become a Real Public Health Problem," in which they estimated that 820,000 deaths per year in Europe and the United States stem from sun deficiency.

What about skin cancer? That's an important question to ask. There are 5 million cases of skin cancer each year in the United States—more than for all other types of cancer combined—and the campaign to make the public more aware of the connection between sunlight and skin cancer has been wildly successful. But over the years, as rates of skin cancer continued to rise despite massive sun-protection efforts, what started as an attempt to prevent overexposure morphed into a kind of fundamentalism. Protect yourself when you're at the beach. No, cover yourself anytime you'll be out and about. Actually, wear sunscreen like a second skin.

While this has helped to grow sunscreen from a niche product with all the gravitas of surfwax into a $16 billion behemoth, one result of this relentless campaign is that the dangers of sunlight loom larger in the public mind than they do in the actual data. I'd always assumed that skin cancer is one

of the leading killers, but it doesn't even make the top 40. Worldwide, it's responsible for about 130,000 deaths per year, which pales in comparison to the 20 million lives lost each year to cardiovascular disease or the 10 million lost to other forms of cancer.
It would be great if we could wave our wand and make skin cancer go away, but if current recommendations might raise the rates of other, more problematic diseases, they need to be reconsidered in light of everything we now know.

That's especially true for people with darker skin, who almost never get sun-induced skin cancer and yet suffer higher rates of many of the diseases that seem to be alleviated by sunlight. So it's one thing to debate the risks and benefits of sun exposure for the small portion of the global population that has fair skin, but for everyone else there's simply no question: Moderate sun exposure provides tremendous benefits, and one-size-fits-all recommendations seem misguided.

At the same time, we have compelling evidence that indoor life is deeply unhealthy. So whatever the fine points, the prescription is the same: *Go outside.* There's no need to overthink it. Wear a hat, wear sunscreen, do what you gotta do, just get out there.

That's right, don't throw away your sunscreen. Arguments over the benefits of sun exposure often devolve into fights over sunscreen, but the two can be complementary. If sunscreens get people outside and allow them to reap the benefits of wavelengths of light other than UV, that's great. Most sunscreens also allow a smidge of UV to sneak through, and often a smidge is all you need.

That's another important consideration. You are not a cactus. Even the researchers most enthusiastic about the sun's salutary effects recommend small doses, depending on your individual situation. If you have fairly pale skin and live in a sunny place, you need just a few minutes a day at most, and you probably get that without even trying. Conversely, if you have dark skin and live in a low-sun environment, you might benefit from a lot more. In between those extremes lies a huge gradient that is further affected by many other variables, including health, age, gender, diet, season, and more. The important thing to remember is that once you understand a few basic principles, many of the benefits of natural light can be had without significant risks.

The practicalities of all this are explored in my new book, *In Defense of Sunlight*, which tells the fascinating story of our changing attitudes toward the sun, from pillar of health to feared carcinogen. But beyond the practical side, I think there's a spiritual side, too. That might be the most precious thing we've lost as we've come to see the sun as an enemy. It casts a shadow across our entire experience of the outdoors, of nature, of the world we were born into. It can trick us into seeking the illusory safety of indoor environments with unhealthy lighting.

That trend has been underway since the Industrial Revolution, in what amounts to a vast experiment to find out what happens when a species that evolved to live in the elements trades up for an indoor existence centered increasingly on the wrong kinds of rays. It will come as no surprise that the results are concerning, especially in younger generations who know nothing else. We are forgetting the heavens.

What's most ironic is that, in a sense, we *are* the heavens. The difference between you and a doorstop is the energy that animates your body. Without that, no movement, no consciousness, no you. That energy comes from sunlight. Most of it fell to Earth, embedded itself in a plant, and then found its way to you. But some of it found you directly. Either way, we are lumps of mud animated by light and having a conscious experience of the universe, and we depend on constant infusions of light to keep the fun going. The good news is that—as of this writing, at least—sunlight is still free, legal, and widely available. Get it while you can.